Human Detection of AI-Generated Phishing Emails: A User Study

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Draft status: Working manuscript. Bracketed items require confirmation from the study records. No numerical results have been fabricated.

Abstract

This study examines whether users can distinguish AI-generated phishing emails from legitimate email messages. Participants were presented with a mixture of legitimate messages and AI-generated phishing messages and asked to classify them. The study also records participants' confidence in their classifications. The analysis plan uses classification measures, including accuracy and phishing recall, calculated from participant responses and verified message labels. The final sample included [N participants] and [N email-classification observations] after [N] warm-up observations were excluded according to [documented exclusion rule]. The observed accuracy was [value, denominator, and confidence interval], and phishing recall was [value, denominator, and confidence interval]. These results should be inserted only after the real, anonymized survey data have been processed. The present draft does not claim a comparison with automated systems because no baseline-system evaluation is documented in the supplied materials.

Keywords— phishing detection; AI-generated email; user study; human decision-making; confidence; survey

I. Introduction

Phishing messages attempt to persuade recipients to disclose sensitive information, transfer funds, or take other actions that benefit an attacker. Generative language models can produce fluent and contextually plausible text, which may make some traditional textual warning signs less dependable. This motivates studying how people classify suspicious email messages when AI-generated phishing examples are included in the task.

The supplied study materials describe a survey in which participants classified legitimate messages and AI-generated phishing messages. The central research question for this manuscript is: How accurately do participants identify phishing messages in this study, and how confident are they in their classifications? A secondary question is whether confidence differs across response categories, provided that confidence was recorded consistently.

The contribution of the study is an empirical description of participant classification performance and confidence in the specific survey setting. Claims about a broad research gap or superiority/equivalence relative to automated detection systems require supporting literature and an actual baseline experiment; they are therefore not asserted as established findings in this draft.

II. Methodology

A. Study design and task

Participants were shown a mixture of legitimate email messages and AI-generated phishing messages and asked to classify each message. The survey also recorded confidence ratings. The following study details must be completed from the survey instrument and collection records: [recruitment method]; [collection dates]; [number and source of email stimuli]; [AI model(s) and generation procedure, if known]; [participant instructions]; [confidence scale and anchors]; and [consent/ethics procedure].

B. Participants and exclusions

The final analytic sample was [N participants]. A warm-up phase was used, and warm-up observations were excluded from the final analysis. The number of excluded observations was [N], and the exclusion rule was [exact rule/phase marker]. Report participant demographics only in aggregate form and suppress small cells where disclosure could identify an individual. Do not publish raw participant-level responses or direct identifiers.

C. Measures and metrics

Let a phishing message be the positive class. A true positive (TP) is a phishing message correctly classified as phishing; a true negative (TN) is a legitimate message correctly classified as legitimate; a false positive (FP) is a legitimate message classified as phishing; and a false negative (FN) is a phishing message classified as legitimate.

- Accuracy = (TP + TN) / (TP + TN + FP + FN).
- Phishing precision = TP / (TP + FP).
- Phishing recall (sensitivity) = TP / (TP + FN).
- F1-score = 2 × (precision × recall) / (precision + recall).
- Specificity for legitimate messages = TN / (TN + FP).

A metric is undefined when its denominator is zero and should be reported as not available rather than as zero. The manuscript must state whether the unit of analysis is an email classification, a participant, or an email stimulus. These units are not interchangeable. If repeated classifications from the same participant are treated as independent observations, this limitation must be acknowledged.

D. Data processing

Data processing was performed in Python using Pandas. The analysis should be run on the approved anonymized dataset. The previous uploaded notebook created synthetic records and therefore did not provide evidence about the real survey. The corrected notebook removes this simulation and requires explicit selection of the real dataset, mapping of the ground-truth and response columns, and explicit identification of warm-up observations before calculating metrics.

III. Results

[INSERT VERIFIED RESULTS ONLY AFTER RUNNING THE CORRECTED NOTEBOOK ON THE REAL ANONYMIZED DATASET.] Report the number of participants, number of eligible classifications, class distribution, excluded warm-up observations, confusion-matrix counts, accuracy, phishing precision, phishing recall, specificity, and F1-score. Include denominators and confidence intervals where appropriate.

Table I. Classification results — to be populated from the verified analysis output. Do not copy values from the previous demonstration notebook: its four retained rows were artificially generated and all had a phishing ground-truth label.

IV. Discussion

[INTERPRET RESULTS AFTER THEY ARE AVAILABLE.] Discuss which message class was more difficult to identify, whether confidence tracked correctness, and what the confusion matrix indicates. Interpret findings in the context of the actual stimuli and participant sample. Do not claim that grammar or wording cues are ineffective unless the study measured those cues.

V. Limitations

The supplied materials do not specify the sample size, recruitment approach, collection dates, number and provenance of email stimuli, AI generation model or prompts, confidence scale, or detailed exclusion count. These

details must be recovered from the study documentation. A convenience sample, a limited set of messages, repeated observations per participant, or a non-representative stimulus set may limit generalizability. No automated-system baseline is described in the supplied materials, so automated-versus-human performance claims are outside the scope of this draft.

VI. Conclusion

This study is designed to describe human classification of legitimate and AI-generated phishing email messages and participants' confidence. A defensible conclusion can be written only after the real dataset has been analyzed and the sample, stimuli, exclusions, and metric denominators have been verified. [REPLACE THIS SENTENCE WITH A CONCLUSION THAT DIRECTLY ANSWERS THE RESEARCH QUESTION USING THE FINAL RESULTS.]

Acknowledgment

The authors thank the survey participants.

References

[ADD verified scholarly references from the project literature review. The supplied one-page PDF contained no reference list; no citations have been invented in this revision. Every factual claim about prior studies should be supported by an appropriate source.]

Pre-submission items to complete

- Confirm affiliation and author names/ordering.
- Insert sample size, recruitment details, collection dates, stimulus provenance, and confidence-scale description.
- Run the corrected notebook on the real anonymized data and insert verified metrics with denominators.
- Add relevant figures and tables generated from the real analysis.
- Add and cross-check scholarly references; remove unsupported research-gap claims.
- Confirm consent/ethics wording and data-availability statement.